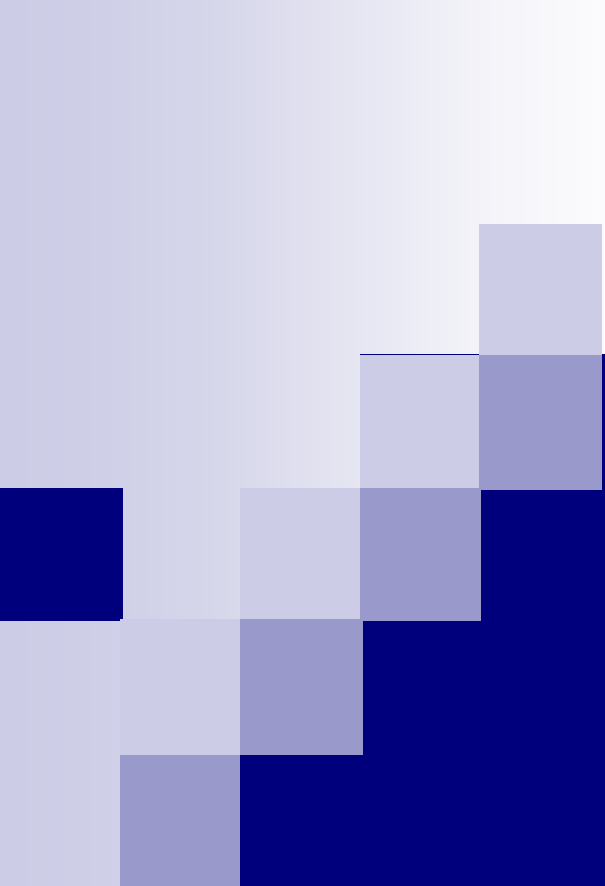




EVOLUTION TRENDS FOR DATABASES

COURSE 13



New types of databases

- NoSQL
- in-memory database (IMDB)
- New SQL



NoSQL databases

- Designed to solve **scalability** and **performance** issues for distributed, cloud-based IT solutions
- Use a different mechanism for storing and accessing data than the one implemented in the relational model, which is based on table structures



NoSQL characteristics

- Allow efficient work with semi-structured and unstructured data
- Use several query languages (NoSQL - not only SQL)
- Do not impose a certain data structure (schema-less)
- Can handle big data volumes
- Use a distributed architecture: data is stored in a redundant way, on multiple servers



NoSQL limitations

- There is no standardization
- There is no efficient data protection
- The consistency of the data is not ensured

NoSQL vs. SQL

	SQL	NoSQL
Data storing	According to relational model, in data tables	Using non-relational specific formats
Data	Structured	Semi-structured, unstructured
Schema	Static	Dynamic
Scalability	Vertical	Horizontal
Query language	SQL	Not only SQL
Joins	Yes	No
Transactions	ACID	CAP



Vertical vs horizontal scaling

Vertical scaling is done by adding more **power** (CPU, RAM) to an existing machine.

Horizontal scaling is done by adding more **machines** into the pool of resources.

ACID transactions:

- Atomicity - multiple operations can be grouped into a single logical entity, that is, other threads of control accessing the database will either see all of the changes or none of the changes.
- Consistency – data is in a consistent state when a transaction starts and when it ends.
- Isolation – how transactional integrity is visible to other users and systems.
- Durability – after a transactions completes, changes to data persist and cannot be undone.

CAP theorem:

- a distributed DBS can only offer 2 of the following 3 guarantees at the same time:
 - Consistency (every read receives the most recent write or an error)
 - Availability (every request receives a (non-error) response, without the guarantee that it contains the most recent write)
 - Partition Tolerance (the system continues to operate despite the arbitrary no. of messages being dropped by the network between nodes)



Data storing in NoSQL databases

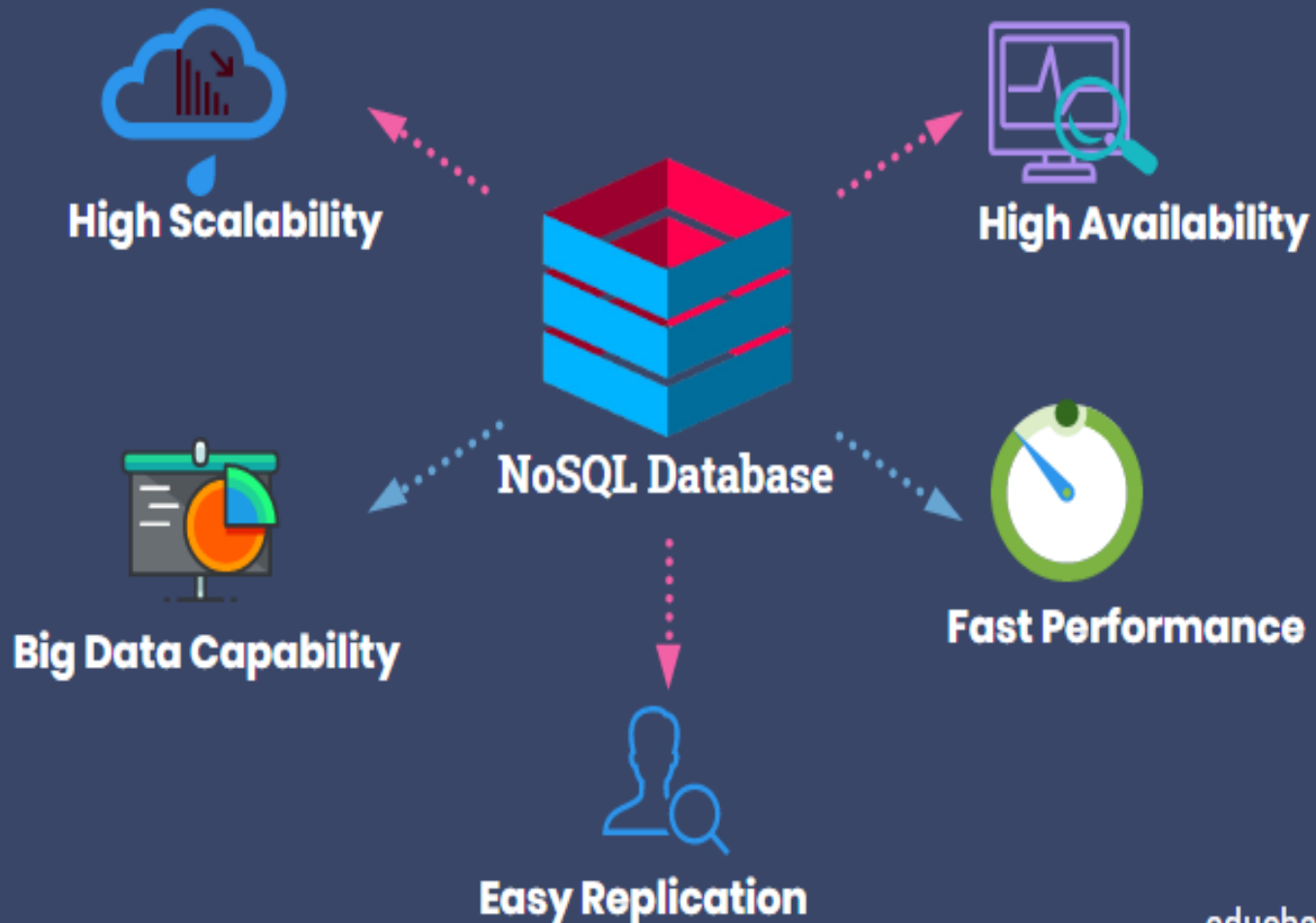
- data is **stored** in cells grouped in **columns** of data
- **document** type storage
- storage based on **key-value structures**
- **graph-based** data storage



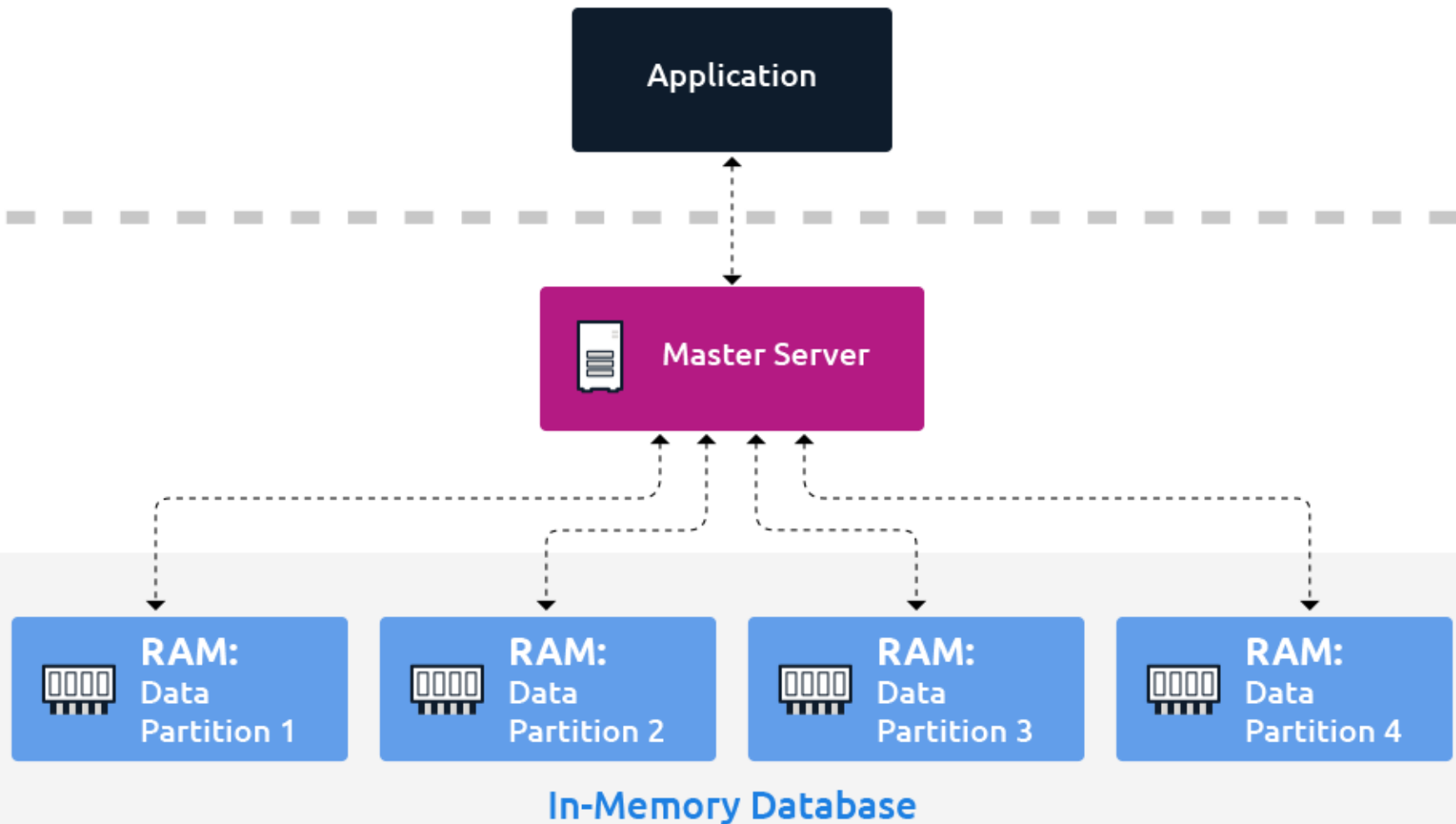
NoSQL DB types

1. Key-value structures DB
(*Redis, Riak, Oracle NoSQL*)
2. Column DB
(*Cassandra, HBase*)
3. Document DB
(*MongoDB, CouchDB*)
4. Graph DB
(*Neo4J, Allegro, Polyglot*)

What is NoSQL Database



<https://hazelcast.com/glossary/in-memory-database/>





in-memory DB

It is primarily based on internal memory for data storage, in contrast to traditional systems that use disk storage mechanisms.

IMDB Characteristics

- Speed - memory access is faster than disk access
- Internal optimization algorithms are simpler
- Executes fewer processor instructions.
- Accessing data in memory eliminates search time when querying data, which offers faster and more predictable performance than disk storage.

IMDB Aplicability

- Applications for which response time is critical:
 - Telecommunications
 - Banking industry
 - Mobile advertising networks
- Data analysis - with the widespread expansion of multi-core processors.



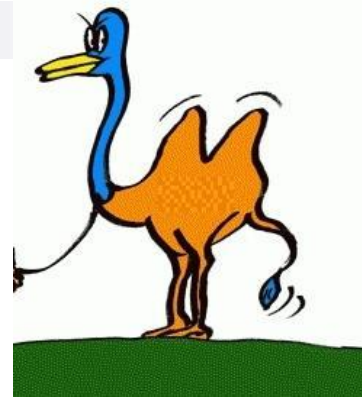
IMDB Limitations

- The volatility of RAM on which data storage is performed
- Data sustainability is not ensured

IMDB examples

DB system	Developer	Description
Apache Ignite (2014)	Apache Software Foundation	in-memory computing platform with powerful SQL, key-value and processing APIs, scales horizontally and provides ACID transactions
RDBMS Oracle 12c (2014)	Oracle Corporation	contains an option for in-memory technology
Microsoft SQL Server 2017	Microsoft	"Memory Optimized Tables" were added to SQL
SAP HANA (2012)	SAP SE	'High Performance Analytic Appliance' is an in-memory, column-oriented, relational database management system
Aerospike DBS (2012)	Aerospike Company	flash-optimized in-memory open source NoSQL database
Redis (2009)	Redis Labs	implements data structure servers, it is networked, in-memory, and stores keys with optional durability
Tarantool (2010)	Mail.ru Group	in-memory database and application server (data grid)

New SQL databases



- A type of *relational database management systems* that attempts to provide the *scalability of NoSQL* systems for online transaction processing (OLTP) tasks, while maintaining the *ACID properties* of a traditional database system.
- Some of the most popular NewSQL DBs are in-memory DBs.



New SQL characteristics

- relational data model and ACID transactional consistency of traditional operational databases;
- SQL familiarity and interactivity;
- NoSQL scalability and speed.



New SQL benefits

- Minimizing application complexity, stronger consistency and often complete transaction support.
- SQL language and standardized tools.
- More complex analytical analysis using SQL and extensions.
- Many systems offer NoSQL-style clustering along with traditional data and query patterns.



New SQL limitations

- No NewSQL system is as general as traditional SQL systems.
- In-memory architectures may be inadequate for data volumes larger than several terabytes.
- Provides only partial access to traditional SQL systems tools.

New SQL examples

- Clustrix,
- NuoDB,
- CockroachDB,
- Pivotal GemFire XD,
- Altibase,
- MemSQL,
- VoltDB,
- c-treeACE,
- Percona TokuBD,
- Apache Trafodion,
- TIBCO ActiveSpaces,
- ActorDB

Conclusion

Characteristic	RDBMS (OldSQL)	NOSQL	NewSQL
ACID transactions	Yes	No	Yes
SQL support	Yes	No	Yes
Standardized	Yes	No	No
Scaling	Vertical	Horizontal	Horizontal
High availability	No	Yes	Yes